cs5800-hw$_{1c}oding_analysis$

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1 Coding and Analysis

1. fib(5):
Recursive Time taken: 0.0 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

fib(10):
Recursive Time taken: 0.0 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

fib(15):
Recursive Time taken: 0.0 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

fib(20):
Recursive Time taken: 0.0 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

fib(25):
Recursive Time taken: 0.01 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

fib(30):
Recursive Time taken in Python: 0.1 seconds

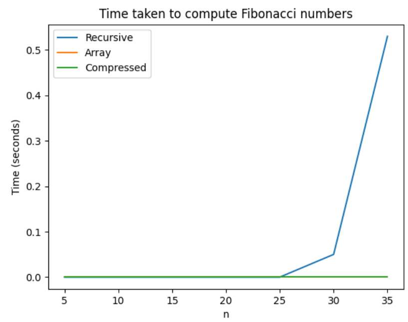

Figure 1: Fib time graphic for Q1

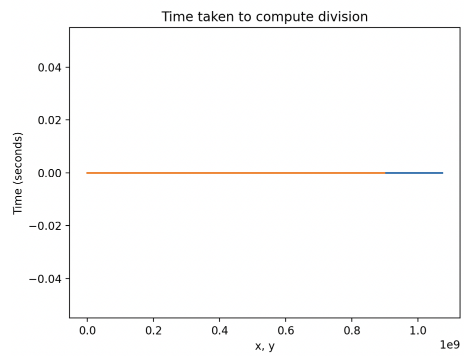

Figure 2: Divide time graphic for Q2

Recursive Time taken in C: 0.06 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

fib(35):
Recursive Time taken in Python: 0.99 seconds
Recursive Time taken in C: 7.38 seconds
Array Time taken: 0.0 seconds
Compressed Time taken: 0.0 seconds

2. Correctness: x=33554432, y=25904487 Divide function result: 1, 7649945 Built-in function result: 1, 7649945

x=67108864, y=41624518 Divide function result: 1, 25484346 Built-in function result: 1, 25484346

x=134217728, y=71080041 Divide function result: 1, 63137687 Built-in function result: 1, 63137687

x=268435456, y=104024179 Divide function result: 2, 60387098 Built-in function result: 2, 60387098

x=536870912, y=347135057 Divide function result: 1, 189735855 Built-in function result: 1, 189735855

Timing: almost constant time.

2 LLM Disclosure

Use LLM to study how to use time.h in C and how to use matplotlib.pyplot to plot the graph.